

System of Linear Equations

Eigen Vectors
Eigen Values

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★ Why we want to solve system of Linear Equations?

(Not writing detailed as not important)

⇒ Predicting Amazon Sales (Using ML which uses system of L.E.)

⇒ Price Prediction (House Price Example)

⇒ Image Classification (cat/dog example)

★ Matrix form of Equations (Obvious Thing)

$$\begin{cases} x + 2y + 3z = 6 \\ 2x + 5y + 2z = 4 \\ 6x - 3y + z = 2 \end{cases} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

$A \quad x = b$

Retail Multiply

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{matrix} 4 \text{ times first col} + \\ 1 \text{ time third col} \end{matrix}$$

→ Here A is coeff. matrix & x is var. matrix & b is solution.

Skip to
the good
part

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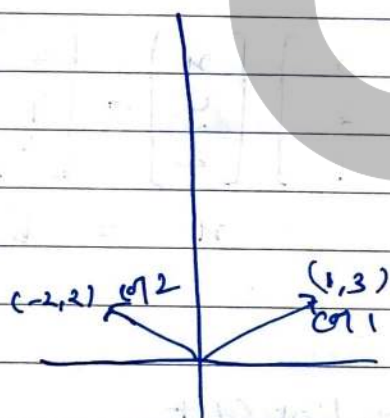
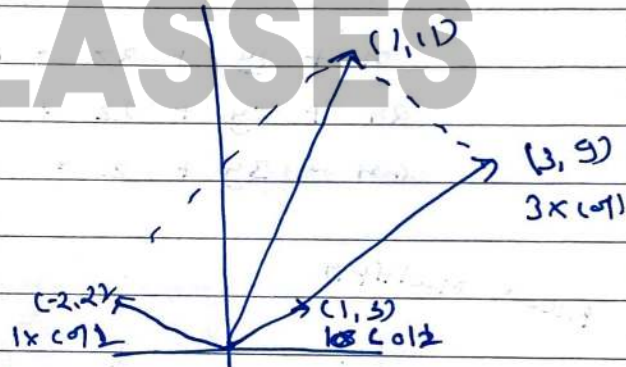
Not Req. just for sake of
completion★ Geometric Interpretation (A little tough)

$$\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

1) Column Picture

$$x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$\text{col}^1 \quad \text{col}^2$

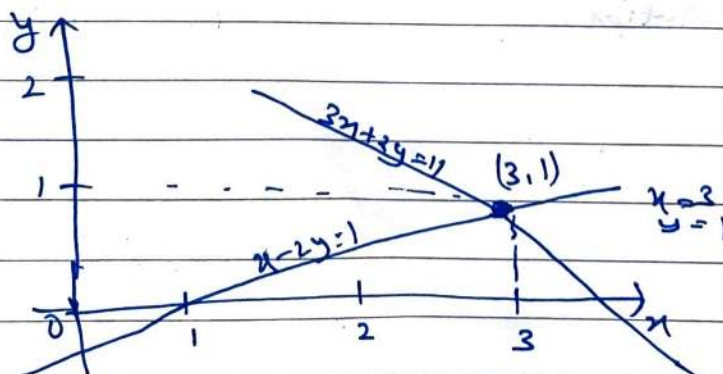
3 times col^1
→ 1 times col^2 →

Column Picture.

2) Row Picture.

$$x - 2y = 1$$

$$3x + 2y = 11$$



Row Picture.

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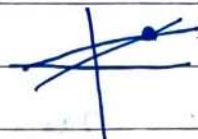
→ It may look like Row Picture is easy, but it is tedious to imagine for more than 2D, so we will move to the column picture.

★ Do we always have soln of $Ax=b$?

Case 1: Unique Soln

$$x_1 - 2x_2 = -1$$

$$-x_1 + 3x_2 = 3$$



Case 2: No Soln

$$x_1 - 2x_2 = -1$$

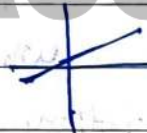
$$-x_1 + 2x_2 = 3$$



Case 3: Infinite Solution

$$x_1 - 2x_2 = -1$$

$$-x_1 + 2x_2 = 1$$



Ques: If $Ax=0$ has some solution, then what can you say about linear dependency of columns of A .

Soln:

$Ax=0$ has some soln means ^{nothing} it is trivial soln. It always have some soln. It is not implying us that any $C_i \neq 0$

$$\text{If } A = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ a_1 & a_2 & a_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} \text{ be } X = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\text{then } c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$$

is there any non zero c_i
 $\downarrow$ No info.

So, nothing can be inferred about C_0 .

Que: If $Ax=0$ has some non trivial solⁿ, then one column of A is L.D.

⇒ Yes, because non trivial solⁿ means we have some $C_i \neq 0$.

so then we can write it in form of L.C. of others, & it makes it L.D.

Que: If $Ax=0$ has only trivial solution then columns of A are L.D.

⇒ False, It is linearly independent as there is no non zero C_i .

Que: Suppose a Matrix $A_{3 \times 3}$ contains 3 L.I. columns. What can you say about no. of solutions to $Ax=b$ ($b \neq 0$)?

⇒ This will be able to cover whole $\mathbb{R}^3$ as we have 3 L.I. vectors, so it will fill the space & so we can generate any 'b'. So inf

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Ans: Same as prev here $A_{3 \times 4}$

$$\Rightarrow \left[\begin{array}{cccc} \text{col}_1 & \text{col}_2 & \text{col}_3 & \text{col}_4 \end{array} \right] \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$\underbrace{\qquad\qquad\qquad}_{\in R^3} \quad \underbrace{\qquad\qquad\qquad}_{\in R^3} \quad \underbrace{\qquad\qquad\qquad}_{\in R^3} \quad \underbrace{\qquad\qquad\qquad}_{\in R^3}$

Solution will always exist as any 3 on LR.Ans: for $A_{5 \times 4}$

$$\Rightarrow \left[\begin{array}{cccc} \text{col}_1 & \text{col}_2 & \text{col}_3 & \text{col}_4 \end{array} \right]$$

$\underbrace{\qquad\qquad\qquad}_{\in R^5}$

Solution may exist or may not exist.It will not exist if b is not L.C. of 3 L.D. vectors else it will exist.(In prev. questions, b was always L.C. of columns as there were n L.R. vectors of R^n)Ans: For $A_{5 \times 2}$, 2 L.R. columns, Solution? $\Rightarrow$ Same Answer. (May or May be not)

Can fill space

Always S_{∞}^n

can't fill space



may or may not

Question: Matrix $A_{2 \times 5}$ contains 3 LI columns.

⇒ Not Possible. to have 3 LI in $A_{2 \times 5}$

Question: Consider a matrix A with dimensions $m \times n$.
For system $Ax=b$, if $m < n$ then it
always have soln? (T/F)

⇒ False. (As it is not filling spaces as
no info about LI columns).

Question: If $m < n$, then there exist sys system
 $Ax=b$ which has soln even if b is
non zero vector? (T/F)

⇒ True, take $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and b as column 1 of A .

Question: If $m > n$ then none of the system $Ax=b$
has soln (T/F)?

⇒ False, may have.

Ques 2: For $m < n$, there is a matrix A which
gives solution of $Ax=b$ for any given b .

⇒ True (Just have m LI columns from $\mathbb{R}^m$).

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Que: for ~~$m < n$~~ $m > n$, there is a matrix A which given solⁿ of $Ax = b$ for any given b .

$\Rightarrow$ False.

Que: for $m > n$ - - - - -

$\Rightarrow$ True.

★

Multiplying two matrices:

$$\begin{bmatrix} \leftarrow a_1 \rightarrow \\ \leftarrow a_2 \rightarrow \\ \leftarrow a_3 \rightarrow \end{bmatrix} \times \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ b_1 & b_2 & b_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix}$$

Elementary school thing!

$$(AB)_{12} = (1^{\text{st}} \text{ row of } A) \cdot (2^{\text{nd}} \text{ col of } B)$$

We can also multiply two matrices the way we were doing matrix and a vector. This is how

$$\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \Rightarrow 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 0 \end{bmatrix},$$

$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

r_1 is linear Comb of b_1, b_2

r_2 is L.C. of b_2

$$\Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 7 \\ 1 & 2 \end{bmatrix}$$

$r_1 \quad r_2$ Ans

Generic Repr.

$$\begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a \begin{bmatrix} p \\ r \end{bmatrix} + c \begin{bmatrix} q \\ s \end{bmatrix} & b \begin{bmatrix} p \\ r \end{bmatrix} + d \begin{bmatrix} q \\ s \end{bmatrix} \end{bmatrix}$$

We will use this interpretation.

Que: If A is an $m \times n$ matrix & B is an $n \times p$ matrix whose first column contains only zeroes, then first column of AB also contains zeroes.

$\Rightarrow$ True, as given

$$\begin{bmatrix} a_1 & a_2 & \dots & a_n \\ b_1 & b_2 & \dots & b_n \\ \vdots & \vdots & \ddots & \vdots \\ c_m & c_m & \dots & c_m \end{bmatrix}_{m \times n} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n \times p} =$$

then first col of AB is

$$0 \times \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ c_m \end{bmatrix} + 0 \times \begin{bmatrix} a_2 \\ b_2 \\ \vdots \\ c_m \end{bmatrix} - - - + 0 \times \begin{bmatrix} a_n \\ b_n \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Imp.
★

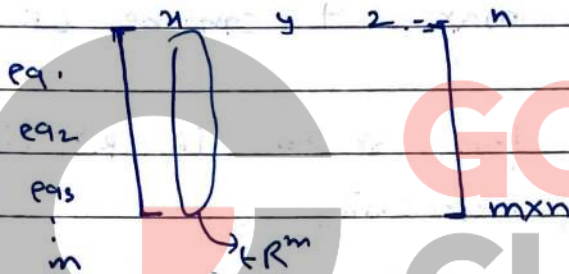
If we have n LI vectors in A then AB can have atmost n LI vectors.

2016
GATE CSEQue:Consider the systems, each consisting of m linear equations in n variables.

- a) if $\frac{m \leq n}{m \leq n}$, then all systems have a solution.
- b) if $\frac{m > n}{m > n}$, then none of " " " "
- c) if $\frac{m = n}{m = n}$, then there exist " " " "

Which statements are correct?

⇒ We can infer the matrix as:



a) False, let $m=2, n=3$. So we have 2 vectors of R^3 , but they may or may not fill the space. So can't say about all systems.

b) False, if $m=2, n=2$ then we have 2 vectors of R^2 , then we can't say that none has solution, there can be a soln.

c) True. there is a system which has a soln.

So only c) is True.

T/F

Ques: a) Every system of 3 eq. in 8 unknown has a solution. (T/F)?

⇒ False, not every system (because nothing is told about LR, so can't fill space).

b) Every set of five vectors in $\mathbb{R}^4$ is LD

⇒ True, in $\mathbb{R}^4$ max. 4 can be LR.

c) Every set of four vectors in $\mathbb{R}^4$ is LD

⇒ False, Not necessarily.

Imp d) If A & B are $m \times n$ matrices such that B can be obtained from A by column operations, then A can be obtained from B by column operations.

⇒ False, Imagine A has vectors $a_1, 2a_1$ & a_2 & B has $a_1, 2a_1, 1$ then we can't get a_2 from B .

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ R_1 & R_2 & R_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}$$

A. $b_1 \quad b_2 \quad b_3$

↓
L.C. of A with b_1 as coeff.
L.C. of A with b_2 as coeff.

★ If b is not a L.C. of cols. get a solution. (Does not matter how hard you try).

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2014
GATE CSE

Que: How many independent columns are present in A ?

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} \begin{bmatrix} 1 & 9 & 5 \end{bmatrix}$$

Soln:

One ^{→ obviously} as

$$\begin{bmatrix} 1 \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} & 9 \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} & 5 \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} \end{bmatrix} \text{ will be the final } A.$$

Que: If b is L.C. of columns of A , then:

- A) There is always a solution of $Ax=b$.
B) There is never a solution of $Ax=b$.

Soln $\Rightarrow$ A is True, B is False. (Think)

Que: A matrix has m LI columns then:

- A) There is always a solution of $Ax=b$.
B) There is never a solution of $Ax=b$ for any b .
C) There may or may not be a solution of $Ax=b$.

Soln: $\Rightarrow$ A is true as we are able to fill the space.

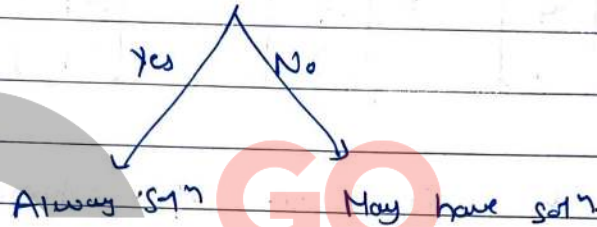
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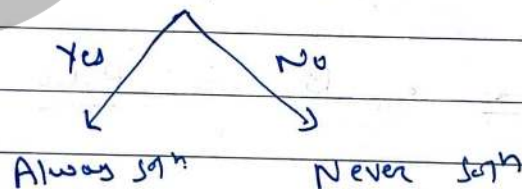
Question: $A_{m \times n}$ doesn't have m linearly dep. then

- A) Always have solⁿ
 B) Never " "
 C) may " "

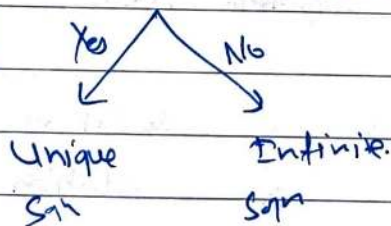
→ $A_{m \times n}$ has m L.C. columns of n ^{variable} ~~columns~~



→ b is L.C. of columns of A



(columns over L.C.)



or vice versa
(Not told till now)

why?

→ Use Brain

⇒ If a vector can be represented as L.C. of few vectors further, if those few vectors are L.I. then there is just one way to do it, otherwise there are many ways.

→ Proof is given later (Mathematic) , but we intuition and think that if col are L.D. and $a_1 = 2a_2$, then obviously we can have another soln by using c_2 instead of c_1 . But if they are L.I., then there will only be one way.

Que: If the linear eq. $Ax=b$ has a unique soln, then the columns of A are L.I.

⇒ True.

Que: How many solutions will the linear system $Ax=b$ have:

a) If b is in the col space of A & the C.I. of columns of A are L.I. → Unique.

b) If b is not L.C. of columns of A . → Not Exst.

2017
GATE

Que: Let $c_1, \dots, c_n$ be the scalars, not all zero such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are col vectors in $\mathbb{R}^n$

Consider the set of linear equations $Ax=b$ where $A = [a_1, a_2, \dots, a_n]$ & $b = \sum_{i=1}^n a_i$. The set of eq. has:

- a) a unique solution at $x=j^n$ where J^n denotes a n -dimensional vector of all 1.
- b) No solⁿ
- c) Infinitely Many solⁿ
- d) finitely many solⁿ

$\Rightarrow$ First line indirectly says that columns are LD. so infinite solⁿ.

Mathematics ↑ Intuition ↓ now

★ Solving S.O.L.E high school way!

$$\begin{array}{rcl} x - 2y & = & 1 \quad \text{--- ①} \\ 3x + 2y & = & 11 \quad \text{--- ②} \end{array}$$

Add ① & ②

$$\begin{array}{rcl} x - 2y & = & 1 \\ 3x + 2y & = & 11 \\ + & & \\ \hline 4x & & = 12 \end{array}$$

$$x = 3 \quad \Rightarrow \quad \text{put } x=3 \text{ in ①}$$

$$3 - 2y = 1$$

$$\begin{array}{l} -2y = -2 \\ y = +1 \end{array}$$

$$\text{So } x=3, y=1$$

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So, we eliminated 'y' & got 'x' & then put 'x' in eq. and got 'y'.

→ Now in case of three eq. and three unknown

$$2x + 4y - 2z = 2 \quad \text{--- (I)}$$

$$4x + 9y - 3z = 8 \quad \text{--- (II)}$$

$$-2x - 3y + 2z = 10 \quad \text{--- (III)}$$

Tricky, but still can be done

Step 1: (II) - 2 * (I) - (III)

2: (I) + (III)

get 'y' & 'z' from (IV) & (V) & then 'x'.

→ If 10000000 eq. and 10000000 unknown, then?

So, basically we have a formal procedure for this called Gaussian Elimination.

★ Gaussian Elimination: Algorithm to solve S.O.E.

Matrix.

Gaussian Elimination →

echelon form of Matrix

Provides Better visualization of Matrix

Leading Entry :

last non zero val
non zero val of row

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* Echelon Form (Or Row Echelon Form)

- ① → All non zero rows are above any rows of all zeroes
- ② → All entries in a column below a leading entry are zero.
- ③ → The leading entry of any row occurs to the right of the leading entry of the row above it.

Ex:

$$\begin{bmatrix} 5 & 1 & -6 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Circles represent leading entries in row.

Shortcut to check all the three rules is by seeing if all leading entries are making stairs or not.

Ex. Not in Echelon Form

$$\begin{bmatrix} 1 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Rule ③, ①, ② failed.

Generic Echelon Matrix

$$\begin{bmatrix} 0 & * & * & * & * & * & * \\ 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & * & * & * \\ 1 & 1 & 1 & 1 & 1 & * & * \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Leading Entries

* → Non zero
⊙ → Anything
0 → zero

Basic $\leftrightarrow$ Pivot
C1 C1

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Basic/

* Pivot & Free Variables: Only exist in Echelon Matrix

In Row echelon form; A variable whose coefficient is a leading non-zero is called a pivot variable.

Otherwise, the variable is known as a free variable.

$$\begin{bmatrix} 0 & * & \textcircled{*} & \textcircled{*} & \textcircled{*} \\ 0 & 0 & 0 & * & \textcircled{*} \\ 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 Pivot Free Pivot Pivot

→ In above def., variable means column as ~~free~~ each variable is represented as a column.

Que: which are basic variables & which are free variables in given augmented matrix?
 $\swarrow$ RHS included in last C1

a) $\begin{bmatrix} 1 & 4 & 3 & 0 & 0 \\ 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\uparrow$ free $\uparrow$ free
 $\downarrow$ Pivot Basic $\downarrow$ Pivot Basic
 ignore. (b/c augmented)

b) $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\uparrow$ Basic $\uparrow$ free $\uparrow$ Basic
 ignore

→ Coefficient Matrix only contains the coefficients ~~not~~ from LHS whereas Augmented Matrix also contains the RHS of the equation in the last col of Matrix along with coefficients.

Ex:
$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases} \text{ LHS}$$

$$\begin{bmatrix} a & b \\ d & e \end{bmatrix}$$

↳ Coeff. Matrix

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

↳ Augmented Matrix

Imp. ★ Free column is always linearly dependent on pivot columns.

★ Elementary Row Operations: Three f.e. row op. 2

- Swap the rows $(R_i \leftrightarrow R_j)$
- Multiply a row by any non zero scalar. $(R_i \rightarrow (R_i))$
- Add to one row a scalar multiple of another. $(R_i \rightarrow R_i + (R_j))$

Ex:

$$R_1 \leftrightarrow R_2 \quad \checkmark$$

$$R_1 \rightarrow R_2 \quad \times$$

$$R_1 \rightarrow R_1 + R_2 \quad \checkmark$$

$$R_1 \rightarrow 3R_1 + R_2 + 4R_3 \quad \checkmark$$

$$R_1 \rightarrow 3R_3 + 5R_2 \quad \times$$

$$R_1 \rightarrow 0R_1 + 7R_2 \quad \times$$

∴ R_i should be in row

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→ These Ele. Row Operations are not for determinants, will be taught later (Point to be considered during revision)

Ques: Convert given matrix to Echelon form / Apply Gauss Elimination to given matrix:

a)
$$\begin{bmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 3R_1$
 As our goal is to have zero at a_{21} & a_{31}

$$\begin{bmatrix} 1 & 3 & 1 & 9 \\ 0 & -2 & -2 & -8 \\ 0 & +2 & 2 & 8 \end{bmatrix}$$

$R_3 \rightarrow R_3 + R_2$

$$\begin{bmatrix} 1 & 3 & 1 & 9 \\ 0 & -2 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\downarrow$ P $\downarrow$ P $\downarrow$ P $\downarrow$ P
 Row Echelon Form

This conversion process is known as Gauss elimination

non-zero
Every row have a leading entry

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$$b) \begin{bmatrix} 0 & 0 & -3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 3 & 1 & 1 \\ 9 & 3 & 5 \\ 0 & 0 & -3 \end{bmatrix}$$

$$\xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{bmatrix} 3 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -3 \end{bmatrix} \xrightarrow{R_3 \rightarrow 2R_3 + 3R_2}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Pivot Free Pivot

★ Rank: Property of Matrix given by any one of these (# → Number of)

- # LI Rows
- # LI Columns
- # Pivot Elements in Echelon form of Matrix.
- # Non zero rows of an Echelon form of Matrix

⇒ These four are always equal.

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Ques:

The rank of Matrix

$$\begin{bmatrix} 0 & 0 & 3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix}$$

is:

- a) 0
- b) 1
- ☒ c) 2
- d) 3

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⇒ Do the Gaussian Elimination:

$$R_1 \leftrightarrow R_3 \rightarrow \begin{bmatrix} 0 & 0 & -3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix} \quad R_1 \leftrightarrow R_3 \rightarrow \begin{bmatrix} 3 & 1 & 1 \\ 9 & 3 & 5 \\ 0 & 0 & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1 \rightarrow \begin{bmatrix} 3 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

↓ ↓
P P

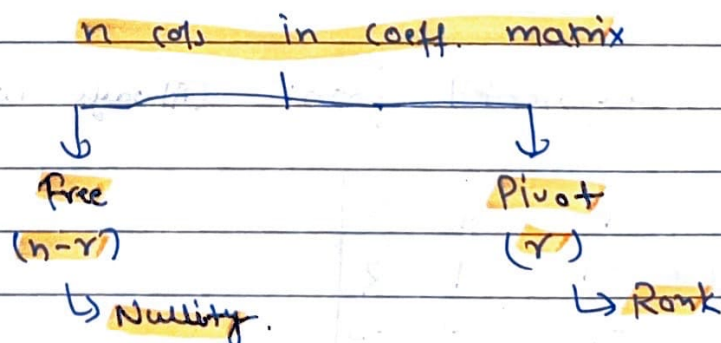
Pivot Elements = 2 ⇒ 2 LP columns as free cols are
LD on pivot elements

Non zero Rows = 2

Ans 2

→ Rank is always positive (≥ 0),
and 0 iff we have zero matrix.

* Pivot Var + Free Var = Total Var = Total No. of Columns



SOLE: System of Linear Equations

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★ Nullity: No. of free columns. (Total col - Pivot col)
Also called the dimension of Null space.

★ Rank Nullity Theorem.

$$\begin{array}{ccccc} \text{Rank} & + & \text{Nullity} & = & n \\ \downarrow \text{Pivot} & & \downarrow \text{free} & & \downarrow \text{No. of columns in} \\ \text{Variables} & & \text{Variables} & & A. \end{array}$$

★ Types of SOLE:

- Homogeneous $AX = 0$
- Non-Homogeneous / Heterogeneous $AX = b$

However, everything will be similar for us for both types.

Ques: Solve the following SOLE using Gaussian Reduction:

$$\text{a) } \begin{cases} x_1 + x_2 - x_3 = 0 \\ x_1 - x_2 + x_3 = 2 \\ 2x_1 - x_2 - x_3 = -3 \end{cases}$$

⇒ In augmented form: (Always write in Aug. form)

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & 2 \\ 2 & -1 & -1 & -3 \end{array} \right]$$

Always form Augmented Matrix

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Now start Gaussian Elimination:

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -2 & 2 & 2 \\ 0 & -3 & 1 & -3 \end{bmatrix}$$

$$R_3 \rightarrow -2R_3 + 3R_2$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -2 & 2 & 2 \\ 0 & 0 & 4 & 12 \end{bmatrix}$$

Now, in Echelon form

Now starting from the last row: (can be written as)

$$4x_3 = 12$$

$$x_3 = 3$$

Now 2nd Row

$$-2x_2 + 2x_3 = 2$$

$$-2x_2 + 6 = 2$$

$$-2x_2 = -4$$

$$x_2 = 2$$

Now 1st Row

$$x_1 + x_2 - x_3 = 0$$

$$x_1 + 2 - 3 = 0$$

$$x_1 - 1 = 0$$

$$x_1 = 1$$

So unknowns are found (i.e., 1, 2, 3) ✓

$$b: \begin{cases} 5x_1 - 11x_2 = -2 \\ -4x_1 + 9x_2 = 1 \\ x_1 - 2x_2 = -1 \end{cases}$$

Aug. Matrix
→

$$\begin{bmatrix} 5 & -11 & -2 \\ -4 & 9 & 1 \\ 1 & -2 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_1 + R_2 - R_3$$

$$R_3 \rightarrow R_1 - 5R_3$$

$$\begin{bmatrix} 5 & -11 & -2 \\ 0 & 0 & 0 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -11 & -2 \\ 0 & -1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{So } -x_2 = 3$$

$$x_2 = -3$$

$$(-4, -3) \checkmark$$

$$5x_1 - 11x_2 = -2$$

$$5x_1 + 33 = -2$$

$$5x_1 = -35$$

$$x_1 = -7$$

c) $\begin{cases} 3x_1 = 6 \\ 2x_2 = 5 \\ \cancel{3x_1} - 4x_3 = 7 \end{cases}$ Already in Echelon form $\Rightarrow$ Directly get the answer.
(Obviously)

$$x_3 = 7/4, \quad x_2 = 5/2, \quad x_1 = 2$$

Imp
d) $\begin{cases} 3x_1 + 5x_2 - 4x_3 = 0 \\ -3x_1 - x_2 + 4x_3 = 0 \\ 6x_1 + x_2 - 8x_3 = 0 \end{cases}$ Aug. $\Rightarrow \begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix}$

$$\begin{array}{l} R_2 \rightarrow R_1 + R_2 \\ R_3 \rightarrow R_3 - 2R_1 \end{array} \Rightarrow \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 3R_2} \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Here x_3 is free variable (Free variable cannot be obtained directly) So in this case:

Step 1. Identify the free var and assign a constant parameter to that (like k, t, p , etc).

So $x_3 = k$.

Now $3x_2 = 0$

$x_2 = 0$.

$3x_1 + 5x_2 - 4x_3 = 0$

$3x_1 - 4k = 0$

$x_1 = 4/3 k$

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Now if $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4/3 k \\ 0 \\ k \end{bmatrix}$

$= k \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}$

Now this eq. has infinite solutions.

if $k=1$, $x_1 = 4/3$, $x_2 = 0$, $x_3 = 1$

if $k=2$, $x_1 = 8/3$, $x_2 = 0$, $x_3 = 2$

so on

This is because, we have one free var, which means that col is LD on other columns, and we already know if we have LD columns then there are infinite solutions.

Imp.

$x - 3y + 2z = 4$

$-x + 2y - 5z = 3$

$5x - 13y + 13z = 8$

$\Rightarrow \begin{bmatrix} 1 & -3 & 1 & 4 \\ -1 & 2 & -5 & 3 \\ 5 & -13 & 13 & 8 \end{bmatrix}$

$R_2 \rightarrow R_1 + R_2$

$R_3 \rightarrow 5R_1 - R_3$

$\begin{bmatrix} 1 & -3 & 1 & 4 \\ 0 & -1 & -4 & 7 \\ 0 & -2 & -8 & 12 \end{bmatrix}$

$R_3 \rightarrow R_3 - 2R_2$

$\begin{bmatrix} 1 & -3 & 1 & 4 \\ 0 & -1 & -4 & 7 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

⇒ for soln do not exist

from last equation

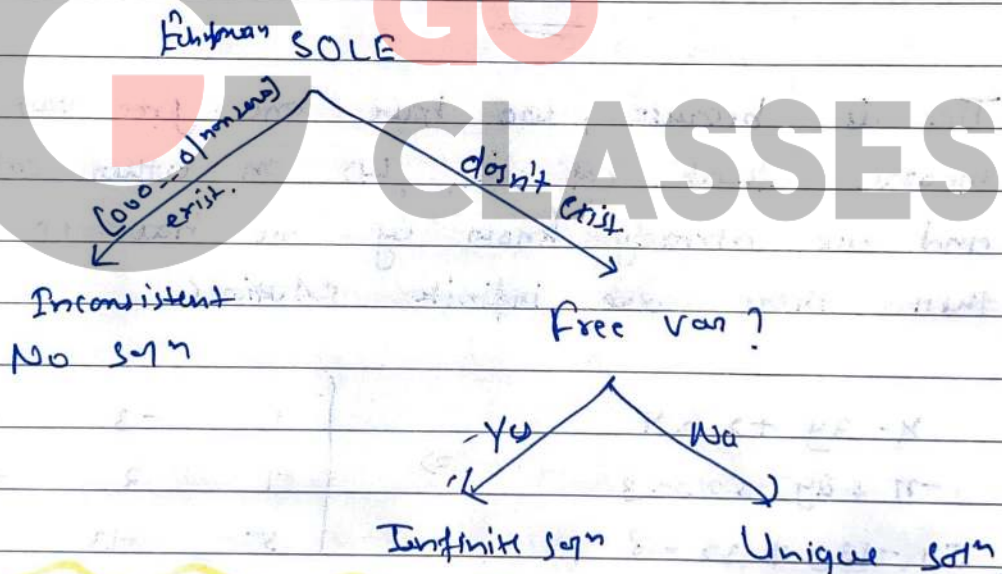
$$0 = 2$$

which is impossible.

So, there is no solution.

So, we have covered unique soln (a, b, c),
infinite soln (d) & no soln (e).

★ Why sometimes soln exists & sometimes not?
→ if we have $[0 \ 0 \ 0 \ \dots \ 0 \ | \ \text{non zero}]$ exist then
no soln else soln.



Ques: A homogeneous system with r eq & s unknown
always have unique soln. (T/F)

⇒ False, it will have soln but unique is
not guaranteed.

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NOTE $\leftrightarrow$ Homogeneous SDE always have a solution, because RHS will be zero, so non zero not possible, which is req. for soln to not exist.

Que: A homogeneous sys. with 10 eq. & 3 unknowns always have unique solution. (T/F)

$\Rightarrow$ False.

Que: Non Homogeneous system doesn't have soln iff $[0 \ 0 \ \dots \ 0 \mid \text{non zero}]$ exist? (T/F)

$\rightarrow$ True that's the only condition.

Que: Find numbers a, b, c, d such that the linear system corresponding to the augmented matrix

$$\begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}$$

has

a) no solutions

$\hookrightarrow d=0, c \neq 0, a \& b$ could be anything

b) Infinite soln

$\hookrightarrow d=0, c=0$

as eq 3 should be free.

Imagine some question solved as (with 2 free var)

$$x = \begin{bmatrix} s+t \\ s-4t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$$

by taking $s=1, t=0$, $t=1, s=0$

And these two vector are always LI.

Here no. of soln : Infinite

no. of LI soln : 2 bcz all soln are
LC of V_1 & V_2 .

Ques: Suppose the parametric form of the solutions to $Ax=0$ looks like :

$$x = s \begin{bmatrix} x \\ y \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} x \\ y \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} x \\ y \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Now, no. of LI columns of A?

$\Rightarrow$ ~~That~~ ~~var~~ ~~2~~ ~~var~~ No. of free var = 3

$$\text{No. of Pivot} = 30 - 3 = 27$$

$$\text{No. of LI columns} = \text{No. of Pivot} = \text{Rank} = \underline{\underline{27}}$$

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Ques: $B = \begin{bmatrix} 1 & -2 & 4 & -3 & 6 & 5 \\ 0 & 0 & 1 & -2 & 0 & -4 \\ 0 & 0 & 0 & 0 & -1 & -5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Solve $Bx = 0$ $\Rightarrow$ It is in echelon form.

So $-x_2 = -5$ $-x_5 - 5x_6 = 0$

$x_2 = 5$ $x_6 = u$, $x_4 = t$, $x_5 = s$

$-x_5 = 5u$

$x_5 = -5u$

$x_3 - 2x_4 - 4x_6 = 0$

$x_3 - 2t - 4u = 0$

$x_3 = 2t + 4u$

$x_1 - 2x_2 + 4x_3 - 3x_4 + 6x_5 + 5x_6 = 0$

$x_1 - 10 + 4(2t + 4u) - 3t + 6(-5u) + 5u$

$= x_1 = -25 + 8t + 16u - 3t + 30u + 5u$

$x_1 = -25 + 5t + 21u - 94$

$x_1 = 25 + 5t + 94$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 25 + 5t + 94 \\ 5 \\ 2t + 4u \\ t \\ -5u \\ u \end{bmatrix} = \begin{bmatrix} 25 \\ 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ 0 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + u \begin{bmatrix} 94 \\ 0 \\ 4 \\ 0 \\ -5 \\ 1 \end{bmatrix}$$

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GATE 2024

Ques

Suppose that P is a 4×5 matrix, such that every soln of the equation $Px=0$ is a scalar multiple of $[2 \ 5 \ 4 \ 3 \ 1]^T$.
The rank of P is _____?

$\Rightarrow$ So basically soln is $s \begin{bmatrix} 2 \\ 5 \\ 4 \\ 3 \\ 1 \end{bmatrix}$ so

free var = 1,

Pivot var = $5 - 1 = 4 = \text{Rank}$

Ans.

Ques:

Suppose a matrix $A_{40 \times 50}$ has 10 pivot columns then -

a) No. of LI solutions in $Ax=0$?

$\Rightarrow$ as. So variables are there

free are 40, i.e. 10 are pivot

So total 40 LI solns are

soln will be in form

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \dots + c_{40} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

b) No. of Linearly Independent Columns in A

$\Rightarrow$ = Rank = Pivot columns = 10

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Ques: Consider a matrix $A \in \mathbb{R}^{m \times n}$. Suppose a system $Ax=0$ has some non trivial solution such that solⁿ are LC of 3 LP vectors. What can you say about.

No. of LP columns.

$\Rightarrow$ Given from above info:

Free vars = 3

So, Pivot vars = $6 - 3 = 3$ = No. of LP columns.

Ques: Let $A = \begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 1 & 0 & 3 \end{bmatrix}$ & $b = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$, solve $Ax=b$.

$\Rightarrow$ Aug. Matrix = $\left[\begin{array}{cccc|c} 2 & 1 & -1 & 2 & 0 \\ 1 & 1 & 0 & 3 & -2 \end{array} \right]$

$R_2 \rightarrow 2R_2 - R_1 = \left[\begin{array}{cccc|c} 2 & 1 & -1 & 2 & 0 \\ 3 & 1 & -1 & 2 & 0 \\ 0 & -1 & 1 & 4 & -4 \end{array} \right]$

free: z, w

$z = s$

$w = t$

$\Rightarrow y + s + 4t = -4$

$y = -s - 4t - 4$ - (2)

$2x + 4 - s - 4t - 5 + 2t = 0$

$x = s + 2t - 1$ - (1)

$$\begin{bmatrix} s & t \\ -s & 4t \\ s \\ t \end{bmatrix} \rightarrow V_r$$

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$$\text{So } \begin{bmatrix} 2+s+t \\ -4-s-4t \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$$

$s=0, t=0$

$s=1, t=0$

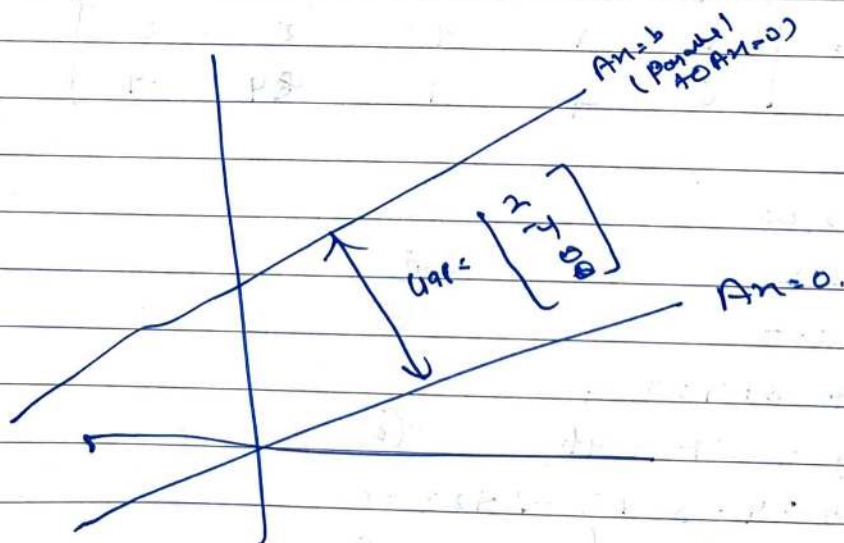
$s=0, t=1$

↑ do not consider 2 & -4

$$\begin{bmatrix} 2 \\ -4 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{t=s} \begin{bmatrix} 3 \\ -5 \\ 1 \\ 0 \end{bmatrix} \xrightarrow{+t} \begin{bmatrix} 3 \\ -8 \\ 0 \\ 1 \end{bmatrix}$$

It was $Ax=b$, but for $Ax=0$ we would have just got V_1 not $\begin{bmatrix} 2 \\ -4 \\ 0 \\ 0 \end{bmatrix}$ constant vector.

Diff. is because



So, Homogeneous will not have constant vector, but heterogeneous will have constant vector.

A $\rightarrow$ Coeff Matrix
A|b $\rightarrow$ Aug. Matrix.

Ques: Rank(A) $\neq$ Rank(A|b), then system has:

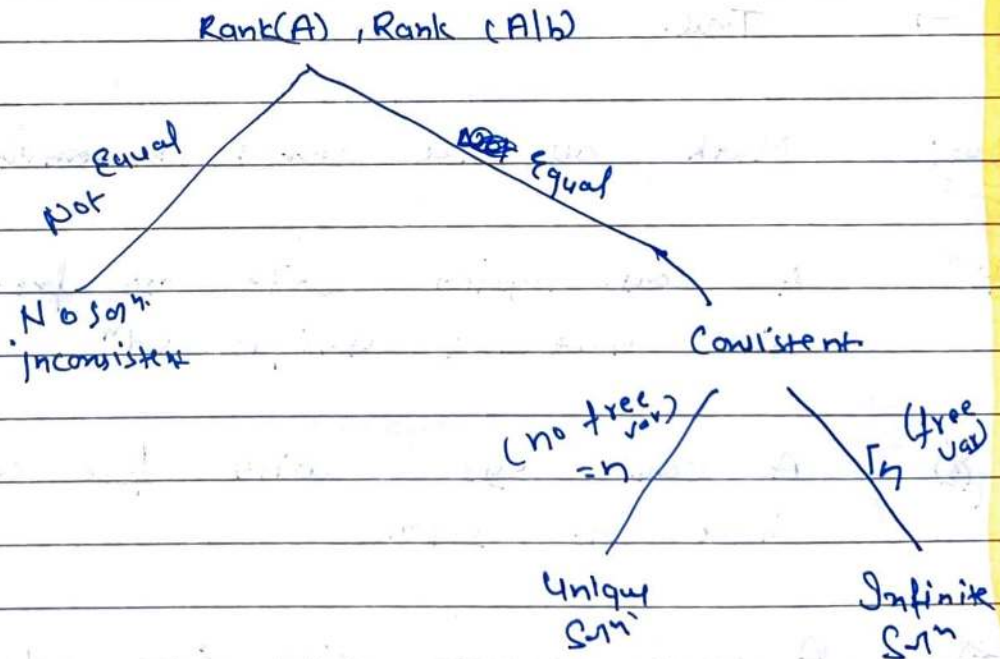
- a) No Solⁿ
- b) Unique Solⁿ
- c) Infinite Solⁿ.

$\Rightarrow$ No Solution. (Use intuition that if rank is unequal then obviously last row is [0 0 0 | nonzero] which is representing no solution)
 $\hookrightarrow$ Non-zero Rows in A & A|b are not equal ✓

Also,

Rank(A) = Rank(A|b) $\Rightarrow$ b is L.O. of A

If Rank(A) = Rank(A|b) then system has unique or infinite solⁿ.



$\hookrightarrow$ n is no. of column

In Consistent $\rightarrow [0 \dots 0 | b]$ exists. $\rightarrow$ No soln

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Ques:

Consider a 3×3 Matrix, $A \in \mathbb{R}^{3 \times 3}$ which of these are possible no. of free columns in A.

☒ 0☒ 1☒ 2☐ 3 $\rightarrow$ zero MatrixQues:

Consider the S.O.M.E in a var with $m \times n$.

I. $Ax=b$ always have a soln where b is a non-zero vector

$\rightarrow$ False. (may not have)

II. $Ax=b$ have a soln only if $b=0$

$\rightarrow$ False.

III. There exists a system which does not have soln.

$\rightarrow$ True.

Ques:

Mark all the correct statements.

☒

A Cons. system with no free col. will have a unique soln.

☒

A Cons. sys. with free cols. will have no soln.

☒

A Cons. system with free cols will have infinite soln.

☒

A incons. sys. with free col. will have infinite solution.

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Que.: Which of the following are true?

- ☒ A SLE (Homogeneous) must be consistent.
- ☒ Multiplying a row of an augmented matrix by zero is an acceptable ele. row op.
- ☒ A single linear eq. with 2 or more unknown will have infinitely many solutions.
- ☒ If no. of eq. in SLE exceeds the # unknown, then solution must be inconsistent.

G.D./E (Optional)

We know in Echelon Matrix
L.R. Rows = # L.C. Columns

But we are doing row operations & then how we are telling L.C. Columns?

⇒ Because even if we perform ele. row op., then ~~the~~ L.D. relation b/w columns remains the same.

Ex.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix}$$

$R_1 \rightarrow R_1 + R_2$

$$\begin{bmatrix} 3 & 5 & 8 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix}$$

Here

$C_3 \rightarrow C_1 + C_2$

Same L.D. Persists

Still

Still

$R_2 \rightarrow R_3 - R_2$

$$\begin{bmatrix} 3 & 5 & 8 \\ 1 & 1 & 2 \\ 3 & 4 & 7 \end{bmatrix}$$

Row Echelon form $\rightarrow$ Echelon form
 Row Reduced " " $\rightarrow$ Reduced " "

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Free Cols $\rightarrow$ LD $\leftarrow$ ~~Row Echelon~~
 Pivot Cols $\rightarrow$ LI $\leftarrow$ Non Zero Rows.

★

Row Reduced Echelon form (Optional) : We have already

Studied Row Echelon form.

$$A \rightarrow \begin{bmatrix} * & * & * & * & * & * & * \\ 0 & * & * & * & * & * & * \\ 0 & 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row Echelon form

$$\begin{bmatrix} 1 & 0 & * & 0 & * & * & * \\ 0 & 1 & * & 0 & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

* $\rightarrow$ Any thing
 * $\rightarrow$ Non zero

Row Reduced E.F.

$\hookrightarrow$ In this (other than RREF, extra is)!

$\rightarrow$ All pivot are 1. (for this do $R_i \rightarrow \frac{1}{*} R_i$)
 $\hookrightarrow$ All elements below & above pivots are zero (Easy)

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Ex:

$$A = \begin{bmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{bmatrix}$$

A

Same Ele.

Row Trans

$$\begin{bmatrix} \textcircled{1} & 3 & 1 & 9 \\ 0 & \textcircled{-2} & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Row E.F.

$$\begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xleftarrow{R_1 \rightarrow R_1 - 3R_2}$$

Row Reduced EF

$$\begin{bmatrix} 1 & 3 & 1 & 9 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xleftarrow{\text{This should be zero}}$$

 $R_2 \rightarrow \frac{1}{2} R_2$

↳

This is useful because using this we are able to find convenient vectors.

(Check C_1 to C_2) which makes process easy
↳ Pivots

So C_3 can be $-2C_1 + 1C_2$

or C_4 can be $-3C_1 + 4C_2$

as C_3 & C_4 are free i.e. LD.

Also holds in Normal Matrix.

↳ But easily derived from Red. EF.

Ex:

RREF Matrix

a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \checkmark$$

b)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & \textcircled{2} & 0 & 1 \end{bmatrix} \times$$

c)

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \checkmark$$

d)

$$\begin{bmatrix} 0 & 1 & \textcircled{3} & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \times$$

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$$e) \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \checkmark$$

★ Convert Matrix to RREF:

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow[\substack{R_1 \leftrightarrow R_2 \\ R_2 \leftrightarrow R_3}]{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_3} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2/2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \checkmark \text{ RREF}$$

Ques: Consider the matrix $A = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 2 & 4 & -2 \end{bmatrix}$

whose RREF is $\begin{bmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{bmatrix}$

Find $a + b + c + d$.

i) -7

ii) -4

iii) -3

iv) -1

v) 0

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⇒ One Approach is to Convert A into RREF & then solve, but that is lengthy. Other is as: w.k. that:

$$\begin{bmatrix} a \\ c \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Also: } a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

(b.c. L.D. remains preserved)

$$2c = 4, \boxed{c = 2}$$

$$a + 3c = 1$$

$$a + 6 = 1$$

$$\boxed{a = -5}$$

Similarly

$$b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$$2d = -2$$

$$\boxed{d = -1}$$

$$b + 3d = 0$$

$$b - 3 = 0$$

$$\boxed{b = 3}$$

$$\text{So } a + b + c + d \Rightarrow -5 + 3 + 2 - 1 \Rightarrow -1$$

Ans.

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Live Session-4 (Before Part-4 Eval & EV)

→ In

$$\begin{bmatrix} s+t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

→ This & this

are LI? → Yes due to 0 & 0.

Also if we shuffle them, logic remains the same.
 i.e. $s \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

→ Every ^{Non zero} row in echelon form of matrix has a pivot, columns may or may not have pivot.
 Only zero rows do not have pivot.

[Old revision done in this session]

→ In matrix $A_{m \times n}$, then

$$\text{Rank}(A) \leq \min(m, n)$$

Ques: Rank(A) = Rank(A|b) then system has:

⇒ Unique Solⁿ or Infinite Solⁿ. bcz. it will not have No Solⁿ and uniqueness depends upon ~~the~~ if Rank = n. where n is no. of columns.

$$\text{rank}(A) \leq \min(m, n)$$

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Ques. Consider a matrix A of $m \times n$ size. Let $\text{Rank}(A) = m = n$.

No. of solⁿ then?

$\Rightarrow$ One solⁿ bcz all colms are LI (Not zero bcz they are covering the space)

Ques. Consider a matrix A of $m \times n$ size. Let $\text{Rank}(A) = n$.

Which No. of solⁿ then?

$\Rightarrow$ 0 or one bcz either 0 or non zero situation will come else one as all columns are ^{having} pivot.

Ques. Consider a matrix A of $m \times n$ size. Let $\text{Rank}(A) = m$.

No. of solⁿ then?

$\Rightarrow$ 1 or ∞ . Here every row is having pivot, and we do not know about cols. but cols $>$ rows bcz cols $>$ rank(A) and $\text{Rank}(A) = m$ so $n > m$. So $n - m$ cols are free & m cols are LI (m is no. of rows). So no solⁿ is not case. Solⁿ can be unique or infinite. (if $m = n$, unique else, infinite).

Ques. Consider a matrix A of $m \times n$ size. Let $\text{Rank}(A) < m$.

$\text{Rank}(A) < n$, No. of solⁿ then?

$\Rightarrow$ 0 or ∞ It means either all rows nor cols are having pivots (And ~~we~~ wkt only zero rows don't have pivot). So no solⁿ can be there if b have some non-zero. And as we have free cols, ∞ solⁿ can be there.

These notes are the property of GOClasses. Please do not share them without their consent.

For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking [here](#)).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal